LS-CAT Management Board Meeting

Agenda

Call In #: 1-888-850-4523
Passcode: 190580

http://conf1.ls-cat.org
http://conf2.ls-cat.org

All times CDT

Location: Call in to building 436A conference room

2:00	Approval of minutes and agenda (JT)
2:10	LS-CAT Status (KB)
2:30	RAPD: Automated MX Data Processing (Frank Murphy, NE-CAT)
2:50	LS-CAT FY 2018 Budget (KB)
3:00	APS Upgrade Status (KB)
3:05	Management Board Chair (JT)
3:15	“Fall” meeting planning: working agenda
3:30	New Business
	Adjourn

Personnel Changes

- Nancy Brennan has retired as of last week
  - We will miss her.
  - Monica Green, the program administrator for DND-CAT, will take over many of Nancy's duties.

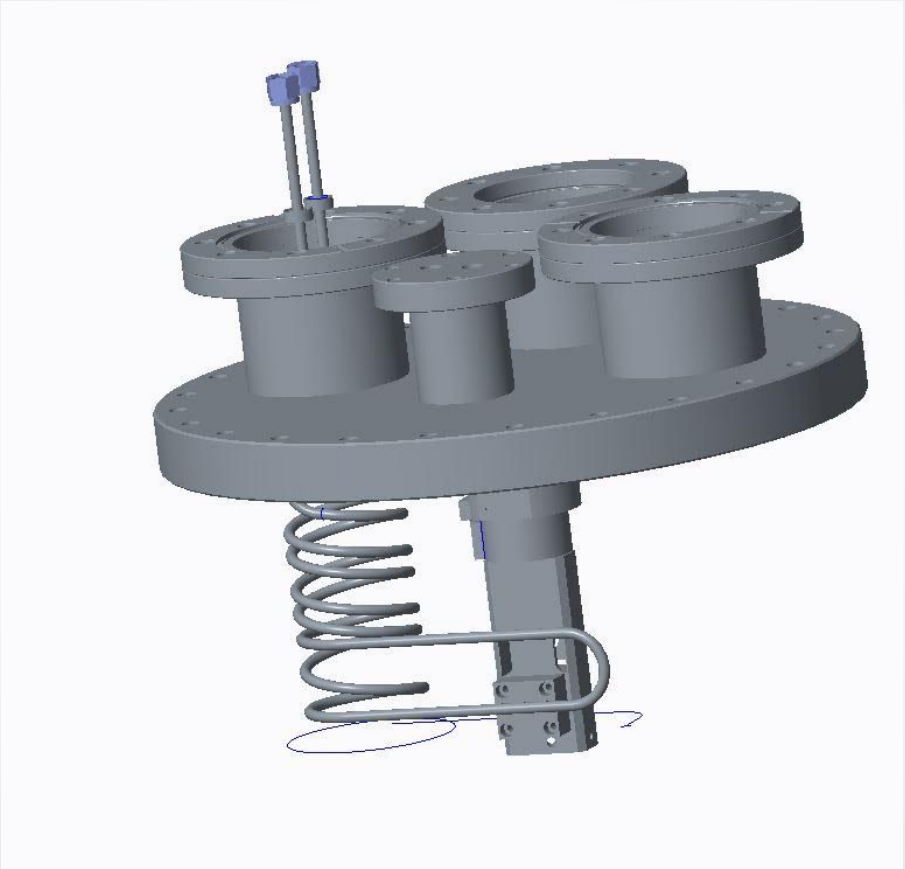
- Reduction in Force: For budgetary reasons we needed to lay off one FTE. David Smith, based on seniority, was selected for this dubious honor.
  - The staff crystallographers will split David's user support duties.
  - Jay VonOsinski and Michael Bolbat will share David's robot maintenance duties.

Side-bounce Mono (F and G) Upgrades

We've had a long standing issue with our side bounce monos. In spite of our requirements document specifying that there be no water to vacuum brazes the delivered produce in fact did contain several such brazes. The manufacturer's solution did not meet our requirements for cooling the diamond crystals so an alternative was developed.

Why now? In order to improve focusing at all of our stations as well as to provide greater flux at low energy on the D station we would like to remove the beryllium window between LS-CAT and the APS storage ring. The APS specifies that we must have no water to vacuum brazes. Removing the Be window requires work by the APS that will become harder to schedule as we approach the APS upgrade.

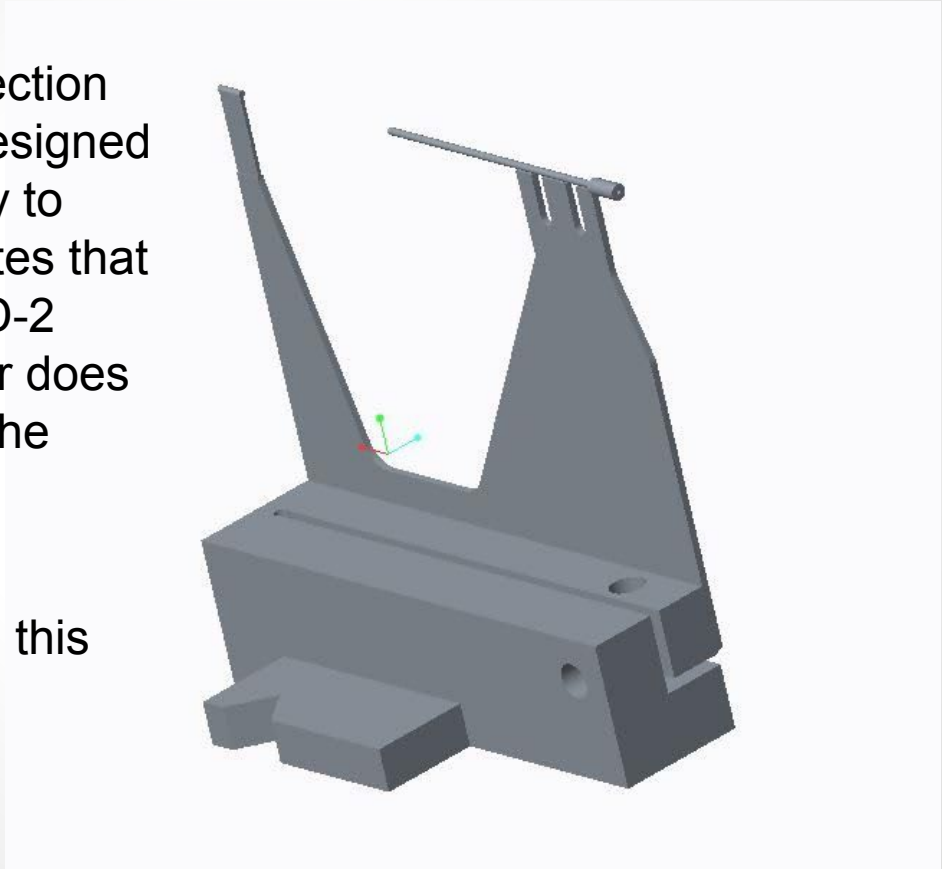
New side-bounce mono cooling scheme.



New beamstop geometry

To support in-situ data collection using micro-plates we've designed a novel beamstop geometry to open up the area of the plates that can be accessed by our MD-2 diffractometers. This holder does not require modification of the existing equipment.

We'll be experimenting with this design over the summer.



Software Updates

Work continues on a replacement for our remote data viewer and it is nearly ready for users to start poking at.

The old viewer is based on Adobe Flash which is rapidly becoming “deprecated”. The new viewer runs in our existing remote interface as an additional tab. We are now installing the additional network and computing infrastructure needed to provide a reasonable response.

Initially we’ll support the Flash interfaces capabilities of viewing, some indexing and data transfers. Once this is stable we’ll be adding other capabilities.



RAPD

LS-CAT is collaborating with NE-CAT on a couple of projects. We provide our diffractometer code and support their NIH proposal. NE-CAT is spending time to set up their Automated MX Data Processing package (RAPD) at LS-CAT.

RAPD
Automated MX Data Processing

Frank Murphy
NE-CAT / Cornell University

Credits

Contributors

Jon Schuermann
David Neau
Kay Perry
Surajit Banerjee
Jim Withrow
NE-CAT Staff

Support

NIH / NIGMS P41 GM103403

Software Dependencies

CCTBX*
BEST
Labelit
Phaser
Pointless
Aimless
CCP4
Mosflm
Phenix
XDS

Why Create RAPD?

Primary Goal: To assist users in making the most of their beam time.

Users are:

1. Overwhelmed by the pace of data collection
2. Not following best practice
3. In need of training on the fly
4. Collecting data remotely

Philosophy of RAPD

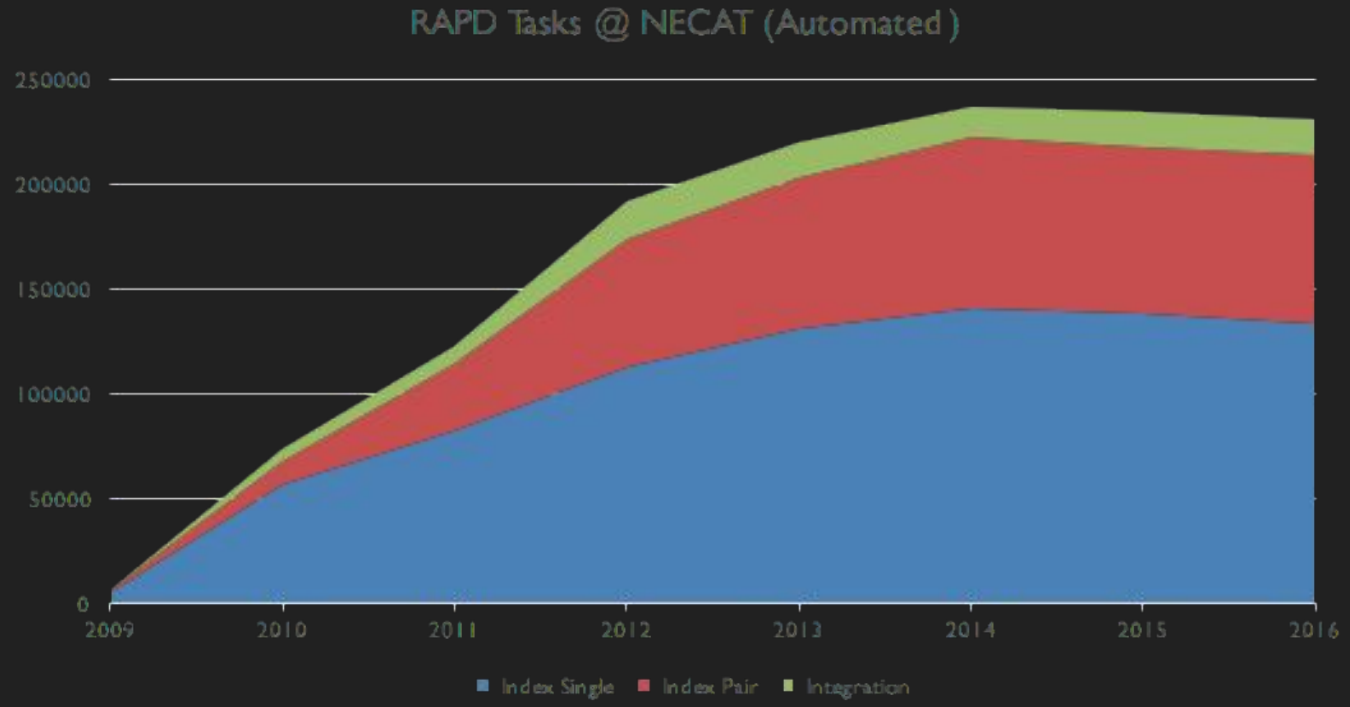
1. Automate repetitive tasks as much as possible
2. Encapsulate beamline support staff knowledge & experience
3. Use best available software
4. Present results as clearly as possible
5. Have reasonable defaults, but as much customization as possible
6. Open source
7. Open to developers

What can RAPD do?

- Index & strategy
- Integration & analysis
- Binary merging*
- SAD*
- MR*

* Not in v2.0 *yet*

RAPD Is Tested



> 1.3 millions tasks performed at NE-CAT

Our Goal for RAPD



Our Result for RAPD



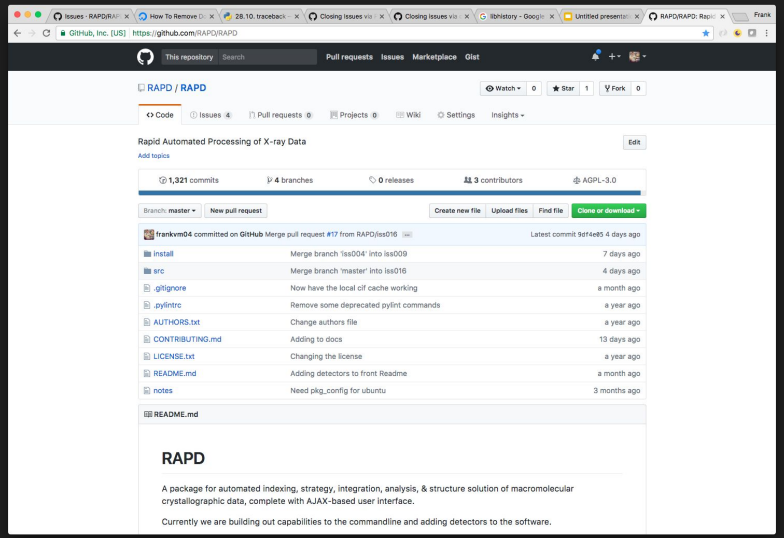
RAPD v2.0 Current Status

1. Building command line tools first
2. Plugins currently available:
    - Index & strategy
    - Integration
    - Analysis
    - PDB search
3. Adding sites proactively and on demand
4. Adding an FTE to support rollout

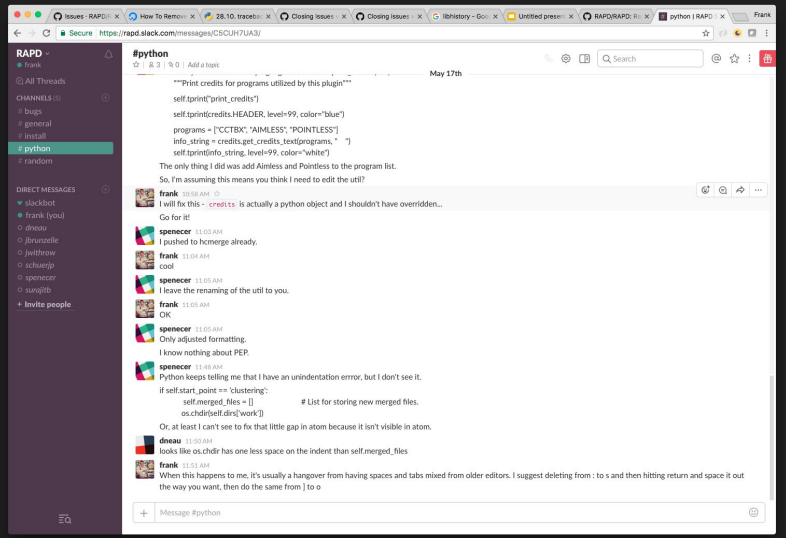




Support



https://github.com/rapd/rapd



https://rapd.slack.com

LS-CAT Budget

Budget Details

FY 17 corrected budget is the approved budget less $123K due to VARI cutting back to ¼ share as of March 1, 2017

FY 17 (Est) is our spending to date plus a guess about the remaining spending. We seem to be living within our means.

Description	FY2017 Budget	FY2017 Corrected	FY2017 Total (Est)	FY 2018 3%
Salaries	911,053	911,053	886,471	798,308
Fringe Benefits	252,362	252,362	241,120	215,543
Supplies	40,000	40,000	23,534	40,000
Supplies, Restricted	2,500	2,500	5,073	2,500
Services	30,000	30,000	27,215	30,000
Services, Restricted	20,000	20,000	28,364	20,000
External Space Rental	123,766	123,766	123,766	127,479
Travel, Domestic	5,000	5,000	7,033	5,000
Utilities & Telephone	9,000	9,000	8,158	9,000
Capital Equipment	96,607	-26,393	92,520	75,910
Indirect Cost - Sponsored	394,666			364,523
Total	1,884,954			1,688,263
Per Share	327,818			337,653

LS-CAT Budget for FY 2018

Salaries			
			100% KB, MB, JVO, SA, JSB, EK, ZW
Base	798,308		50% MAG
Fringe	215,543		
Total		1,013,851	
Items F&A Exempt			
Capital Equip.	75,910		
Rent	127,479		
Telcom	9,000		
Total		212,389	
Non-Salary Items Subject to F&A			
Lab Supplies	40,000		
Office Supplies	2,500		
Lab Services	30,000		
Insurance	20,000		
Travel	5,000		
Total		97,500	
F&A		364,523	
Total		1,688,263	
Per Share		337,653	NU UM MSU
Per 1/2 Share		168,826	UW UI VU
Per 1/4 Share		84,413	WSU VARI

Vote 1:
Approval of amounts shown.

Vote 2:
Exact amount of capital equipment needed depends on what breaks and which upgrades cannot be postponed. Monies left on this line will be added to capital fund or, conversely, when outlays exceed this amount we'll dip into the capital fund. As always, the board will be kept informed about unusually large expenses. Vote 2 allows the capital equipment funds to be used for normal upkeep of the facility.

APS Upgrade Status

- The APS-U has proposed spending some amount (~$20M) on beamline enhancements.
  - We submitted a two proposals, “21-ID-D Replacement Mirrors” and “21-ID-E Fixed Energy Monochromator”
  - It is highly unlikely that these will be funded as BES will not support a life sciences sector. The main value for us to submit them is to have the APS spend some effort to estimate the costs for each.

- Proposed FY18 federal budget includes $40M for the APS-U project
  - This is double previous years and required to start the dark period in 2022
  - Susceptible to politics

- The CD-2 (PDR) is planned for early 2018.
  - This is where we find out what the APS-U is actually planning to include.

LS-CAT Management Board Chair

Meeting in November (or December)

Proposed Fall Meeting Agenda

- Issues that affect the budget to be presented in May.
- More in depth status reports.
- Invite APS and/or APS-U participation.
  - APS updates us on their current status
  - We update the APS on our needs and wants
- Invite University VPR representation
- Encourage in-person participation.
- Venue can be but does not need to be the APS

Proposed May Meeting Agenda

- Status updates.
- Pass the Budget.
- Mainly remote participation (but you are always welcome to come in person).